ACCREDITED STEM PROGRAMME | SAFE 3V BATTERY POWERED | AGES 8–12 | INCLUDES USER MANUAL & DIY PROJECTS

Electronics STEM Kit for Kids (Ages 8–12)

Offered by: **Corvit Systems Rawalpindi** • Lead Facilitators: **Engr. Aamir Jamil & Miss Zia Ul Ain**

Educational Aim & Learning Vision

To make students familiar with fundamental concepts of electronics and circuits, preparing them with confidence for future robotics, automation, and advanced engineering courses.

📘 **Included User Manual:** Comes complete with a comprehensive illustrated guide allowing students to build the 6 core workshop projects plus numerous additional DIY electronics projects at home.

4 Hours	6 Projects	14+ Parts	100% Safe
2 Lessons (2 hrs each)	Hands-on builds	Kit & User Manual	No code / solder / 220V

Lead Facilitators & Trainers (Corvit Systems Rawalpindi)



Engr. Aamir Jamil
Lead STEM & Electronics Trainer
Experienced engineer and robotics mentor specialized in child-safe hardware prototyping and experiential electronics pedagogy.



Miss Zia Ul Ain
Robotics & STEM Educator
Specialist in youth STEM training and lab supervision, empowering learners to grasp circuit polarities and sensory feedback.

Programme Core Objectives

1. Component Recognition & Polarity:
Identify breadboard buses, LED polarity (anode/cathode), resistors, switches, buzzers, sensors, and DC motors.

2. Circuit Physics & Current Flow:
Grasp voltage, current loops, open vs. closed paths, and why current-limiting resistors prevent burnout.

3. Sensors & Energy Conversion:
Convert electrical energy into light, audible sound (buzzers), and physical movement (motors & propellers).

4. Troubleshooting & Independent DIY:
Diagnose loose jumper leads and reverse polarities, empowered by the included manual for home experiments.

Structured 4-Hour Course Syllabus

Designed with a 50/50 balance of intuitive visual theory and hands-on laboratory experimentation.

Lesson 1: Electronics Fundamentals

DURATION: 2 HOURS

Core Topics: Introduction to STEM & circuits; battery DC voltage; conductors vs. insulators; solderless breadboard navigation; LED polarity; resistor values & color codes; manual push-button switching; open and closed circuits; buzzer acoustic frequency.

Timeline	Workshop Activity	Pedagogical Focus
00-15 min	Introduction & electrical safety briefing	Lab rules, low-voltage safety
15-30 min	Component identification & breadboard exploration	Grid mapping & bus lines
30-55 min	**Project 1:** Mini LED Torch construction	Complete loops & LED polarity
55-80 min	**Project 2:** Push-Button Light assembly	Switching & loop interruption
80-105 min	**Project 3:** Electronic Quiz Buzzer panel	Sound wave generation
105-120 min	Testing, circuit diagnostics & assessment	Troubleshooting skills

Lesson 2: Sensors, Alarms, and Motion

DURATION: 2 HOURS

Core Topics: Input sensors vs. output transducers; gravity-activated mechanical tilt switches; Light-Dependent Resistors (LDRs); photo-resistance variance; electro-mechanical energy conversion; DC motor coils & magnetic fields; propeller airflow dynamics.

Timeline	Workshop Activity	Pedagogical Focus
00-15 min	Review of Lesson 1 & sensor/motor introduction	Input vs. Output devices
15-40 min	**Project 4:** Tilt-Activated Alarm calibration	Gravity orientation sensors
40-70 min	**Project 5:** Light Detector (LDR) experiments	Optical resistance modulation
70-100 min	**Project 6:** Mini Motor Fan assembly	Rotational kinetic propulsion
100-115 min	Creative STEM design challenge & speed test	Creative engineering application
115-120 min	Comprehensive review, outcome celebration & feedback	Self-guided DIY transition

Lesson 1: Fundamental Circuit Projects



Engr. Aamir Jamil guiding torch assembly

Project 1: Mini LED Torch

25 MINS • LIGHT OUTPUT

Objective: Build a complete battery-powered handheld torch while mastering LED polarity and current-limiting resistor protection.

Components Used:

2 × AA batteries, battery holder, 1 × bright LED, 1 × 330Ω resistor, jumper wires, cardboard mounting cylinder.

Step-by-Step Procedure:

1. Identify the positive (+) and negative (-) terminals of the battery pack.
2. Identify the LED anode (longer leg, positive) and cathode (shorter leg, negative).
3. Connect battery positive to the 330Ω protective resistor.
4. Wire resistor output to the LED anode and LED cathode back to battery negative.
5. Test illumination and firmly house the circuit into a cardboard torch prototype.

✓ Learning Outcome: Complete loop understanding, LED polarities & resistor safety.



Students testing push-button breadboard loop

Project 2: Push-Button Light

25 MINS • SWITCHING CONTROL

Objective: Control an LED dynamically using an interactive push button to grasp open and closed circuit behavior.

Components Used:

Mini breadboard, tactile push-button switch, 1 × LED, 1 × 330Ω resistor, 2 × AA battery pack, jumper wires.

Step-by-Step Procedure:

1. Replicate the basic LED torch circuit onto the solderless mini breadboard.
2. Place the tactile push-button switch across the center divider of the breadboard.
3. Route the circuit current through the push-button terminals in series.
4. Press the button to close the circuit and illuminate the LED immediately.
5. Release the button to break the circuit (open loop) and extinguish the LED.

✓ Learning Outcome: Physical understanding of electrical switching and current interruption.

Acoustic Signaling & Gravitational Sensing



Engr. Aamir Jamil & students with Quiz Buzzer

Project 3: Electronic Quiz Buzzer

25 MINS • ACOUSTIC SOUND

Objective: Convert electrical energy into audible sound waves for a classroom game quiz contestant panel.

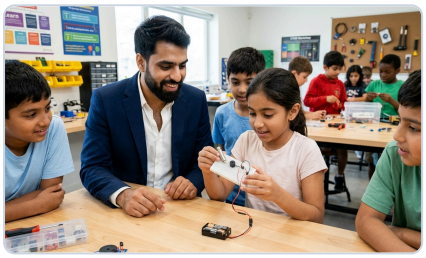
Components Used:

Active piezo buzzer, push button, battery pack, jumper wires, decorative cardboard quiz console.

Step-by-Step Procedure:

1. Verify the active buzzer's rated operating voltage and polarity (+ and - leads).
2. Connect the buzzer and push button in series with the battery power supply.
3. Press the push button to complete the loop and activate the audible tone.
4. Release the button to instantly silence the buzzer.
5. Install into a colorful cardboard contestant buzzer console for competitive quizzes.

✓ Learning Outcome: Electrical-to-acoustic energy conversion & audio signal triggering.



Demonstrating tilt sensor angle & alarm trigger

Project 4: Tilt-Activated Alarm

25 MINS • ORIENTATION SENSOR

Objective: Build a motion-sensitive security alarm using a mechanical gravity-ball tilt switch sensor.

Components Used:

Metallic cylindrical tilt switch, active buzzer, mini breadboard, battery supply, jumper wires.

Step-by-Step Procedure:

1. Mount the cylindrical tilt switch vertically on the breadboard.
2. Wire the tilt switch in series with the active buzzer and battery pack.
3. Observe switch conduction as the internal metallic ball shifts with gravity.
4. Gently tilt the breadboard past 45° to sound the intrusion alarm.
5. Discuss real-world applications such as earthquake alarms and security sensors.

✓ Learning Outcome: Understanding gravitational orientation sensing and security circuits.

Optical Resistance & Electro-Mechanical Motion



Observing photo-resistance change by shading LDR

Project 5: Light Detector

30 MINS • LIGHT SENSING

Objective: Investigate photo-resistance physics by testing how ambient light regulates electron flow through an LDR.

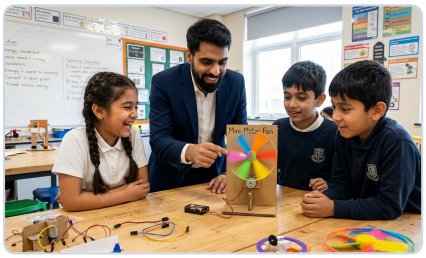
Components Used:

1 × Light-Dependent Resistor (LDR), 1 × LED, 1 × 330Ω resistor, 2 × AA batteries, mini breadboard, wires.

Step-by-Step Procedure:

1. Examine the photo-conductive serpentine track on the LDR sensor face.
2. Construct a series circuit with LDR, protective resistor, LED, and battery pack.
3. Observe LED brightness under direct classroom overhead lighting.
4. Cup hands over the LDR to shade it and observe the sharp change in resistance.
5. Learn how automated streetlights and solar sensors operate on this principle.

✓ Learning Outcome: Variable semiconductor resistance and automated optical detection.



Engr. Aamir Jamil & students with spinning fan

Project 6: Mini Motor Fan

30 MINS • KINETIC MOTION

Objective: Convert electrical energy into rotational kinetic motion and aerodynamic airflow using a 3–6V DC motor.

Components Used:

Small 3–6V DC motor, lightweight plastic propeller blade, cardboard support stand, 2 × AA battery pack, wires.

Step-by-Step Procedure:

1. Inspect the DC motor terminals and verify voltage ratings.
2. Secure the motor firmly to the rigid cardboard support mount.
3. Press-fit the lightweight plastic propeller fan blade onto the motor spindle shaft.
4. Connect motor leads to the battery supply and observe rotation and airflow.
5. Reverse the battery connections to discover how polarity changes spin direction!

✓ Learning Outcome: Electromechanical conversion, aerodynamic propulsion & polarity reversal.

Required Kit Hardware Bill of Materials (14 Items)

Component Name	Educational Function	Qty
Mini Breadboard	Solderless rapid circuit prototyping platform	1
LEDs (Mixed Colours)	Visual output indicators, polarity training (Red/Green/Yellow)	6
Resistors (220Ω, 330Ω, 1kΩ)	Current-limiting safety and circuit protection	10
2 × AA Battery Holder & Cells	Safe low-voltage power supply enclosure with alkaline batteries	1 + 2
Jumper Wires (Male-to-Male)	Flexible plug-and-play circuit interconnects	10
Push Buttons & Crocodile Clips	Tactile loop switching & mechanical electrical gripping	2 + 2
Active Buzzer (Piezo)	Audible acoustic frequency alert signaling	1
Small DC Motor (3–6V) & Propeller	Electro-mechanical kinetic rotation and aerodynamic airflow	1 + 1
Light-Dependent Resistor (LDR)	Optical environmental sensing and photo-resistance	1
Tilt Switch Sensor	Gravitational motion and orientation intrusion detection	1
Detailed Illustrated User Manual	Full-color diagrams, troubleshooting & bonus home DIY project blueprints	1

📘 User Manual & Multi-Project Potential:

The kit includes an illustrated guidebook enabling students to independently build the 6 workshop projects plus numerous home DIY projects (e.g. Morse code transmitters, multi-LED logic circuits, vibration sensors, and light-sensitive alarms). Modular IR obstacle sensors can be added as an optional curriculum extension.

Safety Guidelines

- Low-voltage 3V battery power only; zero mains hazard.
- Disconnect battery prior to altering circuit leads.
- Always protect LEDs with current-limiting resistors.
- Secure motor mounts before applying electrical power.
- Continuous expert adult supervision throughout.

Assessment & Takeaways

- Sketch and build functional breadboard loops.
- Explain LED polarity and resistor protection.
- Calibrate tilt switches and optical LDR sensors.
- Diagnose and resolve real circuit faults.
- Prepared for robotics & micro-controller courses.

Corvit Systems Rawalpindi — Executive Authorization
Pioneers in Technical Education & Advanced STEM Labs in Pakistan

Instructors: Engr. Aamir Jamil & Miss Zia Ul Ain
✓ Certified STEM Proposal